

Module Plan for Sequence Learning

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Key Topics to Include: (Brainstorming)

- **Introduction to Sequence Data:** Characteristics of temporal and sequential data (time series, text, audio); why standard feedforward networks fail here.
- **Recurrent Neural Networks (RNNs):** Unrolling through time; hidden state updates; Backpropagation Through Time (BPTT).
- **The Gradient Problem:** Mathematical intuition behind vanishing and exploding gradients in deep/unrolled networks; gradient clipping.
- **Gated Architectures:** Long Short-Term Memory (LSTMs) and Gated Recurrent Units (GRUs); cell states, forget gates, input/output gates.
- **Sequence-to-Sequence (Seq2Seq) Models:** Encoder-Decoder architectures for variable-length inputs and outputs (e.g., machine translation).
- **Decoding Strategies:** Greedy search vs. Beam search; temperature scaling in text generation.
- **Bridge to Modern Architectures:** Limitations of Seq2Seq bottlenecks; conceptual introduction to Attention mechanisms.

Online Learning Roadmap

1. Sequential Data
 - a. [Sequential and Time Series Data Explained | by Ahamed Shahmi | Medium](#)
 - b.  3_Recurrent Neural Networks & Sequential Learning
2. RNNs and Gated Architecture, and Gradient Problem
 - a.  3_Recurrent Neural Networks & Sequential Learning
 - b.  Recurrent Neural Networks (RNNs), Clearly Explained!!!
 - c.  Long Short-Term Memory (LSTM), Clearly Explained
3. Sequence to Sequence Models
 - a. [Understanding how a Seq2Seq Model works for Machine Translation: Detailed Explanation of each step | by Abhinav Battu | Medium](#)
 - b.  4_Sequence-to-Sequence Learning
 - c. [Deep Learning Series 06:- Types of Seq2Seq Models | by Yashwanth S | Medium](#)
 - d. Notebook: [seq2seq.ipynb](#)

4. Decoding Strategies

- a. [Understanding greedy search and beam search | by Jessica López Espejel | Medium](#)
 - b.  11 Beam Search Decoding
 - c.  4_Sequence-to-Sequence Learning
- #### 5. Bridge to Modern Architecture (Transformers)
- a.  12 Attention Mechanisms Part 1
 - b. [Attention is All you Need](#)